

# Correctness Scores Alone Are Unreliable Safety Certificates

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## Abstract

Process reward models (PRMs) score the local correctness of reasoning steps, making them natural candidates for deciding whether a step is safe to roll back and regenerate. We test whether a correctness score can *license* such a safety decision—not whether reasoning can be compressed in general. Using a counterfactual protocol (control, deletion, a position- and length-matched *placebo* deletion with its own control, and a meaning-preserving *paraphrase*) on 600 rating-balanced PRM800K steps, with replications on two further generators, a within-checkpoint thinking-mode toggle, and 300 ProcessBench steps, we find that deletion gains concentrate almost entirely in steps that are both low-rated and semantically harmful (+23.3 pp; difference-in-differences +25.4 pp): erroneous steps act as *negative anchors* for no-thinking generation. An apparent generic “restart” benefit disappears once each placebo is matched to its own control, and paraphrasing the erroneous step performs like keeping it; toggling the same generator into extended thinking makes the anchor vanish. Turned into rollback policies under a natural-prevalence weighting with nested train/calibrate/test selection, correctness-based certificates are split-unstable and checkpoint-dependent: one trained PRM certifies in every split yet meets its budget on only 9 of 20 test splits, while another transfers on most—so correctness scores alone cannot be trusted as safety certificates a priori. A single probe-and-rollback generation removes all detectable excess harm in a gold-verifier simulation. Correctness scores nominate rollback candidates; by themselves they are unreliable safety certificates. We release the audited dataset, pipeline, and benchmark.

## 1 Introduction

Process reward models assign scores to individual steps of a reasoning trajectory and have become a standard tool for verifying and steering

large language model (LLM) reasoning (Lightman et al., 2024; Uesato et al., 2022; Wang et al., 2024). Work on reasoning compression motivates asking whether an intermediate step can be safely altered, but compression quality is not our estimand. Our narrower question is whether a PRM correctness score can safely trigger a counterfactual rollback. Reusing correctness scores as rollback signals embeds a silent assumption: that a step’s local correctness tells us what happens to downstream reasoning if the step is *removed*. Before that reuse hardens into practice, we test the assumption directly.

This assumption conflates distinct properties of a step: it can be locally correct yet contribute nothing later steps use; locally incorrect yet arrive after the trajectory is already doomed; or incorrect in a way that actively *anchors* the continuation to a bad path, so removing it rescues the run. Whether rollback is safe is a *counterfactual* property of the step-generator pair, not a syntactic property of the step in isolation.

**Main question and answer.** Can PRM scores safely decide which reasoning steps to remove? We find that correctness scores alone cannot: they detect local wrongness, but do not determine semantic contribution or counterfactual removability. Safe rollback therefore requires counterfactual validation rather than a correctness threshold alone.

We study this gap directly. Starting from PRM800K (Lightman et al., 2024), we sample 600 target steps balanced across human ratings (−1/0/+1), annotate each step’s semantic contribution (Essential / Redundant / Harmful), and measure what actually happens when a fixed generator (Qwen3-8B; Qwen Team, 2025) continues the trajectory with and without the step, four runs per condition, 2,400 paired outcomes. Because deleting *any* step also forces the generator to re-plan—which can itself change outcomes—we additionally run a position-and-length-matched *placebo*

deletion for 511 of the 600 steps (1,514 runs), decomposing raw deletion effects into target-specific and generic-restart components. An LLM judge scores final answers, audited against 200 human adjudications (headline sensitivity  $\leq 0.13$  pp).

Three findings answer this question. **Correctness is not removability.** Correctness and semantic contribution are correlated but far from interchangeable, and deletion gains concentrate overwhelmingly in steps that are both low-rated and Harmful (+23.3 pp), while rating-0, rating-+1, Essential, and Redundant steps are statistically indistinguishable from zero; the pattern replicates on a second no-thinking generator and on external ProcessBench trajectories. **The gains are negative anchors, not restarts.** A single-control placebo suggests a generic “restart” benefit, but that contrast is confounded; giving each placebo cutoff its own control shows innocuous deletions are inert and the anchor effect is fully target-specific, and paraphrasing the erroneous step—preserving its (wrong) meaning—performs like keeping it. The effect is generation-mode-specific: toggling the *same* checkpoint into extended thinking makes the anchor vanish. **Rollback needs counterfactual validation, not a threshold.** Turned into deployment-weighted rollback policies with nested train/calibrate/test selection, correctness-based certificates are split-unstable and checkpoint-dependent—one trained PRM certifies in every split yet meets its budget on only 9 of 20 test splits, another transfers on most, and correctness quality does not predict which. A single probe-and-rollback generation closes most of the gap, removing all detectable excess harm in a gold-verifier simulation.

Our contributions follow this main line: (i) we separate PRM correctness, semantic contribution, and retrospective removability and show that correctness does not imply removability; (ii) we provide a counterfactual protocol that isolates negative-anchor mechanisms with matched placebos, own controls, and semantic-preserving paraphrases; and (iii) we turn the construct gap into a rollback risk-coverage benchmark, showing when correctness signals fail to certify safety and how a cheap probe-and-rollback check repairs the risk. We release all data, code, and audits (Appendix C).

The conceptual chain is *PRM rating*  $\rightarrow$  (correlated but non-equivalent) *semantic contribution*  $\rightarrow$  (not determined by either) *counterfactual removability*. Figure 1 puts this distinction and the intervention safety test before the detailed results.

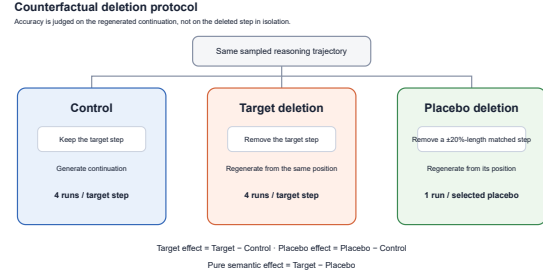


Figure 1: Study logic and counterfactual protocol. PRM rating (local correctness) is correlated with, but not equivalent to, semantic contribution; counterfactual removability is a further layer measured by regenerated Control, Target-deletion, and matched Placebo continuations. A correctness rating may nominate a candidate but is not itself a rollback-safety certificate.

## 2 Related Work

**Process supervision and PRMs.** Step-level supervision improves credit assignment over outcome-only rewards (Uesato et al., 2022; Lightman et al., 2024; Cobbe et al., 2021). Math-Shepherd derives step labels from rollout success probabilities (Wang et al., 2024); Setlur et al. (2025) argue for rewarding *progress* (prospective advantage) rather than local correctness; generative PRMs verify steps with explicit reasoning (Khalifa et al., 2025). Benchmarks probe what PRMs detect: ProcessBench locates first errors (Zheng et al., 2025), and PRMBench separates fine-grained error dimensions including redundancy (Song et al., 2025). None measures what happens when a step is *removed*—the counterfactual layer our protocol supplies.

**Reasoning interventions and rollback safety.** Token- and step-pruning methods study efficiency-oriented alterations to chains of thought using attention-based redundancy (Choi et al., 2025), step entropy (Li et al., 2025), or likelihood-preserving greedy deletion (Singh and Hakkani-Tür, 2026). We do not evaluate compression efficiency or claim that reasoning cannot be compressed. Instead, we test the narrower safety question raised when PRM correctness is reused as a rollback trigger: does local correctness certify the downstream safety of removing a step? These works primarily preserve the answer of an already-correct trajectory; we additionally quantify the *repair* regime and the safety of the intervention itself.

**Deletion as intervention.** Ablation-style causal analyses of reasoning typically mask content at the token level or measure answer-probability shifts. Our design instead runs full *regenerated* continuations—the deployment-relevant intervention—and controls the regeneration with matched placebos that carry their *own* controls, which proves essential: a shared-control placebo manufactures an apparent +10 pp “restart” component that disappears, and understates the semantic effect, once each cutoff is read against its own control (Section 6).

### 3 Constructs and Estimands

We distinguish four properties of a step  $s_t$  in trajectory  $h$ : (1) *local correctness*, the PRM800K rating in  $\{-1, 0, 1\}$ ; (2) *semantic contribution*, whether the content of  $s_t$  is needed by (Essential), irrelevant to (Redundant), or misleading for (Harmful) downstream reasoning; (3) *prospective advantage*, the change in success probability from *taking* the step (Setlur et al., 2025); (4) *retrospective removability*, the change in success probability from *deleting* the step after the fact and regenerating. This paper measures (4) and its relation to (1) and (2).

For step  $i$ , let  $Y_{ij}^c, Y_{ij}^t \in \{0, 1\}$  be final-answer correctness under **Control** (step kept) and **Target** deletion, with  $K=4$  independent runs per condition,  $\bar{Y}$  denoting means over runs and  $p_i^c, p_i^t$  the underlying success probabilities; matched steps (511/600) additionally receive one to four **Placebo** runs  $Y_{ij}^p$  (1,514 in total), applying the same truncation operator at a different, length-matched step  $p$ , and every placebo cutoff receives an *own control*  $C_p$  (prefix kept through  $p$ ; 1,513 of 1,514 scheduled runs returned). Pre-specified estimands (fixed in the released analysis plan before the extension cohorts ran; we did not use an external registry):

$$\begin{aligned} \Delta_i^{\text{tgt}} &= \bar{Y}_i^t - \bar{Y}_i^c, & \Delta_i^{\text{plc}} &= \bar{Y}_i^p - \bar{Y}_i^c, \\ \Delta_i^{\text{sem}} &= \Delta_i^{\text{tgt}} - \Delta_i^{\text{plc}}, \end{aligned} \quad (1)$$

plus the own-control difference-in-differences  $\Delta_{\text{DiD},i}^{\text{sem}} = \Delta_i^{\text{tgt}} - (\bar{Y}_i^p - \bar{Y}_i^{C_p})$ , adopted after the shared-control confound in  $\Delta^{\text{plc}}$  was discovered (Section 6; the pre-specified contrast is retained as *legacy* in Table 14, appendix), plus run-level *discordance* events *danger* ( $Y^c=1, Y^t=0$  on independent draws) and *benefit* ( $Y^c=0, Y^t=1$ ). Runs are independent fresh-session samples with no shared randomness across conditions; pairing control run

$j$  with deletion run  $j$  is a fixed bookkeeping convention whose expected discordance rate equals  $p_i^c(1-p_i^t)$  under any pairing, so danger/benefit rates estimate those products—independent-sample discordance, never within-seed counterfactual flips (Section 9 builds the causal safety metric). Discordance is reported either as a joint rate (share of all pairs) or conditioned on the control draw’s outcome; we state which at each use. All confidence intervals are 95% percentile intervals from 5,000-replicate cluster bootstraps over *target steps*; re-clustering by problem leaves headline intervals essentially unchanged (Table 14, appendix). Fixed seeds are released.

### 4 Data and Intervention Protocol

**Step cohort.** From PRM800K phase-2 test trajectories (Lightman et al., 2024) over MATH problems (Hendrycks et al., 2021) we sample 600 target steps, 200 per rating, balanced over early/middle/late positions (201/201/198). **Unless otherwise stated, all main results use the frozen AI-assisted semantic labels; human-calibrated labels are only used for explicitly identified robustness or diagnostic validation.** Semantic contribution was annotated twice, with both passes completed before any deletion outcome existed: an initial AI-assisted pass (165 Essential / 201 Redundant / 234 Harmful) and a human-calibrated re-labeling pass (198 / 199 / 203); the passes disagree on 164/600 steps (27.3%), an explicit reliability bound; a blinded four-annotator pilot provides a human-only robustness check (Table 6; full agreement details in Appendix B). The frozen statistical pipeline behind Tables 1–2 uses the *initial* pass as its primary label set ( $-1 \times$  Harmful anchor: 178 steps); the calibrated pass (anchor: 160 steps, 154 shared) is confined to the correctness-vs-contribution diagnostic, extension-cohort sampling metadata, and explicitly labeled robustness re-estimations (Table 14, appendix; the headline survives either pass). Under the calibrated labels, rating and type are associated (Cramér’s  $V = 0.528$ ) with 65.5% of steps on the expected diagonal—correlated, but with a third of the cohort off-diagonal (Figure 3, appendix).

**Human-only label pilot.** The blinded four-annotator pilot (120 steps, blinded to ratings, prior labels, and outcomes; pooled  $\kappa = 0.40$ ) gives a human-only check on the semantic taxonomy: on its 15 exact-consensus Harmful steps the target-deletion effect is +25.0 pp, reproducing the anchor



pattern from labels assigned without outcome access (Table 6, appendix). We read this as a sensitivity analysis, not a full-cohort reannotation.

**The anchor effect is not a position, length, or label-frequency artifact.** Target-deletion gains are positive in all three balanced position bins, and the  $-1 \times \text{Harmful}$  anchor effect is positive in every bin (Tables 7 and 10, appendix). Adding log target length to a problem-clustered regression leaves a small, non-significant length coefficient (+1.2 pp per log unit; Table 11, appendix). Permuting the semantic labels within their observed margins (5,000 draws) centers the within-rating Harmful contrast at zero, while the observed label-rating intersection (26.7%) far exceeds the permutation null (9.5–13.0%; Table 8, appendix).

**Interventions.** Both interventions are prefix truncations of the same trajectory: the Control prompt contains the problem plus steps  $1:t$ , the Target-deletion prompt contains steps  $1:t-1$ —operationally a *rollback* of the most recent visible step, not an interior splice—and Qwen3-8B (temperature 0.7, top- $p$  0.8) regenerates the continuation, four fresh runs per condition. The Placebo condition applies the identical truncation at a different step  $p$  of the same trajectory (before or after  $t$ ; 51%/49% in practice) whose token length is within  $\pm 20\%$  of the target’s, choosing the closest available position; 511/600 steps admit such a match (1,514 placebo runs), and the 89 unmatched steps differ from matched ones only modestly (largest imbalance: position SMD 0.298). The placebo-cutoff steps are benign: none carries rating  $-1$  (Table 15, appendix).

**Outcome evaluation and audit.** An LLM judge (Qwen3-8B, temperature 0; prompts released) marks final-answer correctness against ground truth. We audited 200 stratified outputs with human adjudication: agreement 93.0%, maximum per-condition bias 2.3 pp—below the pre-set gates (90%/5 pp; detail in Table 4, appendix). Substituting all 200 adjudicated labels into the frozen tables moves the overall target effect by  $-0.08$  pp and the placebo-corrected effect by  $-0.05$  pp—observed sensitivity on the audited subset, not a bound on the remainder; no conclusion moves under substitution (Table 4). Because the judge shares a checkpoint with the primary generator, a deterministic PRM800K-style normalization+SymPy grader independently regrades all 6,314 primary

runs: 93.0% agreement, headline effects within 0.9 pp (Appendix A).

**Replication and mechanism cohorts.** Four extensions reuse the identical prompt protocol. (i) *Cross-generator*: 300 steps (100 rating  $-1 \times \text{Harmful}$  anchors, 100 rating-0, 100 rating- $+1$ ; 103 stably-correct; positions balanced) rerun with **phi-4** (14B), 3 runs per condition, reusing the frozen matched-placebo indices; the identical 2,691-task list is additionally rerun with **DeepSeek-R1-Distill-Qwen-14B** (long-CoT, served locally; sampling and judge unchanged) and—holding the checkpoint fixed—with **Qwen3-8B’s own thinking mode enabled** (same provider, prompts, sampling; 3,582 runs incl. 891 own-controls).<sup>1</sup> (ii) *Semantic-preserving paraphrase*: for a 240-step strong-control subset (80 anchors / 80 stably-correct / 80 fragile-neutral, rating 0/ $+1$ ), an independent model (Gemini 2.5 Flash) paraphrases the target step preserving its exact meaning *including errors*; 221/240 paraphrases passed automatic validity checks (7.9% excluded) and each ran 4 continuations, alongside 4 fresh matched-placebo runs. Condition codes: C0 = frozen control, C1 = frozen target deletion, C2 = fresh matched placebo, C3 = fresh paraphrase. (iii) *Deployed PRM scoring*: five PRMs score all 600 steps—three trained checkpoints (Qwen2.5-Math-PRM-7B; Math-Shepherd; RLHFlow-Llama3.1-8B) and two prompted LLM-judge PRMs (Gemini 2.5 Flash, phi-4). (iv) *External validation*: 300 ProcessBench steps (Zheng et al., 2025) spanning GSM8K, MATH, Olympiad-Bench, and Omni-MATH—a benchmark whose trajectories come from twelve source models; six appear in our sample (Section 7).

## 5 Full-Cohort Deletion Results

**Semantic labels add resolution beyond rating.** Ignoring semantic type and selecting all rating- $-1$  steps gives a  $+22.0$  pp deletion effect; adding the Harmful label concentrates the set to 178 steps at  $+23.3$  pp, while the 22 rating- $-1$  non-Harmful steps show a weaker  $+11.4$  pp (Table 9, appendix). The within-rating contrast is positive but imprecise ( $n=22$ ): semantic labels sharpen subgroup resolution without creating the headline by discarding rating information.

<sup>1</sup>R1-Distill was unavailable via the API provider at freeze time; phi-4 matched the 7–14B no-thinking budget (all 2,691 continuations parsed). The local long-CoT rerun parsed 99.0%; its 28 unparseable continuations, balanced across conditions, score as incorrect.

| Group       | Steps | $\Delta^{\text{tgt}}$ (pp) | 95% CI           |
|-------------|-------|----------------------------|------------------|
| Overall     | 600   | +7.21                      | [+3.88, +10.58]  |
| rating = -1 | 200   | +22.00                     | [+15.41, +28.42] |
| rating = 0  | 200   | +0.25                      | [-5.41, +5.90]   |
| rating = +1 | 200   | -0.63                      | [-5.54, +4.21]   |
| Essential   | 165   | -1.21                      | [-7.24, +4.78]   |
| Redundant   | 201   | +1.12                      | [-3.87, +6.06]   |
| Harmful     | 234   | +18.38                     | [+12.28, +24.44] |
| -1×Harmful  | 178   | +23.31                     | [+16.57, +30.24] |

Table 1: Target-deletion effects on final-answer accuracy (600 steps, 4,800 runs in 2,400 pairs, cluster bootstrap over steps). **All rows in this primary table use the frozen AI-assisted initial-pass semantic labels**; the human-calibrated anchor estimate (+20.9 pp) is a robustness result reported in Table 14 (appendix).

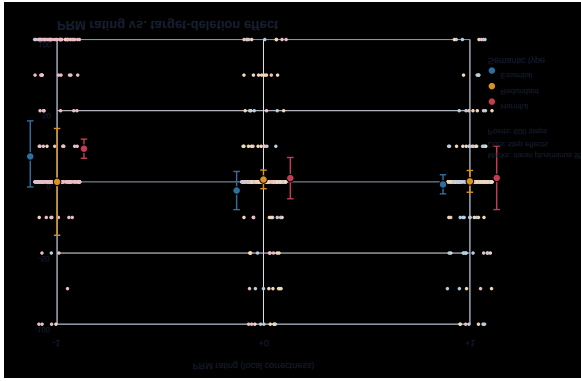


Figure 2: PRM rating versus target-deletion effect on the 600-step primary cohort. Each point is one step; colors show the frozen AI-assisted semantic label. Large positive effects concentrate in rating = -1 Harmful steps, whereas rating-0/+1 groups cluster near zero. Larger markers show within-cell means with 95% step-bootstrap intervals.

Deleting target steps helps on average (+7.21 pp, Table 1), but the average is misleading: the entire reliable gain sits in the rating = -1 column and the Harmful row, peaking at +23.31 pp for their intersection (Figure 2), while the off-diagonal cells are small- $n$  and inconclusive. Correct (rating +1) and Essential steps show near-zero *average* effects—but not because deletion is safe there. Danger discordance is substantial everywhere: among paired draws whose control sample is correct, the deletion sample is wrong 15.1% of the time (the joint rate is 8.5%; neither is an individual causal flip, Section 3), and even within rating = -1 it is 14.9%. The near-zero group means arise from offsetting discordance in both directions, not from inertness.

**Trajectory state dominates raw effects.** Stratifying by control stability makes the state-

dependence explicit (Figure 4, appendix): steps whose four control runs are all wrong gain +33.4 pp from deletion (nothing to lose), while steps whose control runs are all correct *lose* 12.1 pp (CI [-15.5, -8.9]). These strata condition on the same control draws that enter  $\Delta^{\text{tgt}}$ , so regression to the mean pushes the extreme strata outward; we read them as descriptive bounds on state-dependence, not causal subgroup effects—and a deployment policy never observes these counterfactual states, which is why raw deletion gains overstate what a policy can capture (Section 9).

**Step-level stability.** With four paired runs per step, 377/600 steps never change outcome; among low-rated Harmful steps, 29% are strongly beneficial to delete ( $\geq 3$ -run gain) vs. 11% elsewhere—enrichment, not a guarantee.

**Cross-generator replication.** The pattern is not a Qwen3-8B artifact. Rerunning 300 steps with phi-4 reproduces every qualitative conclusion: overall target effect +12.4 pp, anchor-group +33.7 pp with own-controlled placebos inert (-1.3 pp) and a DiD semantic component of +34.9 pp, while rating-0/+1 effects are indistinguishable from zero. Effect *magnitudes* are generator-dependent, and step-level signs agree for 65% of the 80 jointly-nonzero steps—above chance but far from deterministic.

**Extended thinking neutralizes anchors.** Two thinking-mode variants show the anchor is specific to no-thinking generation. Under DeepSeek-R1-Distill-Qwen-14B, which thinks before answering, anchor-step control accuracy rises to 77.7% and the anchor-group deletion gain drops to +7.0 pp—but this is a content-agnostic truncation benefit, not an anchor effect: deleting a matched *innocuous* step helps as much (+8.0 pp own-control) and the DiD semantic component is -1.0 pp (n.s.). Because R1 changes model, scale, and data at once, we also toggle thinking on the *same* Qwen3-8B checkpoint (identical provider, prompts, sampling): on the anchor cell, thinking lifts control accuracy from 26.8% to 83.3% and the deletion effect collapses from +21.8 to -3.0 pp (DiD semantic -3.5 pp, n.s.). The collapse is thus attributable to extended thinking itself, not to the R1 checkpoint.

## 6 Mechanism: Anchors, not Restarts

Does deletion help because the *content* of the step was harmful, or because *any* deletion lets the gen-

| Group             | $\Delta^{\text{tgt}}$ | $\Delta^{\text{plc}}_{\text{vs C}}$ | $\Delta^{\text{plc}}_{\text{own}}$ | $\Delta^{\text{sem}}_{\text{DiD}}$ (CI) |
|-------------------|-----------------------|-------------------------------------|------------------------------------|-----------------------------------------|
| Overall (511)     | +7.97                 | -0.57                               | +0.46                              | +7.52 [+2.8, +12.4]                     |
| rating = -1 (169) | +22.78                | +9.47                               | -0.69                              | +23.47 [+14.4, +32.5]                   |
| rating = 0 (176)  | +0.99                 | -5.63                               | +2.84                              | -1.85 [-9.7, +6.0]                      |
| rating = +1 (166) | +0.30                 | -5.42                               | -0.90                              | +1.20 [-5.9, +8.4]                      |
| Harmful (201)     | +18.41                | +3.57                               | -1.74                              | +20.15 [+12.0, +28.4]                   |
| -1×Harmful (151)  | +23.84                | +9.99                               | -1.60                              | +25.44 [+16.1, +34.9]                   |

Table 2: Placebo decomposition (pp; initial-pass labels).  $\Delta^{\text{plc}}_{\text{vs C}}$ : placebo vs. the *target’s* control (frozen contrast);  $\Delta^{\text{plc}}_{\text{own}}$ : vs. its *own* control  $C_p$  (1,513 new runs);  $\Delta^{\text{sem}}_{\text{DiD}} = \Delta^{\text{tgt}} - \Delta^{\text{plc}}_{\text{own}}$ . Matched innocuous deletions are inert in every group of this cohort.

erator restart and resample? Table 2 answers with two placebo contrasts. Against the target’s control, placebo truncation inside the anchor group looks strongly positive (+9.99 pp; rating -1 overall +9.47)—the reading a single-control design would report as a “restart benefit”. But this contrast confounds removing step  $p$  with moving the cutoff: 51% of matched cutoffs fall *before* the target, so those placebo prefixes silently drop the erroneous step too. Splitting by side is diagnostic: cutoffs before the target yield +21.1 pp against the shared control (they remove the anchor), cutoffs after it yield -6.8 pp (they keep it). Giving each placebo cutoff its *own* control—new  $C_p$  runs that keep step  $p$  and truncate after it—removes the confound:  $\Delta^{\text{plc}}_{\text{own}}$  is null in every group (-1.6 pp on anchors, +0.5 pp overall), and the difference-in-differences semantic effect *rises* to +25.4 pp (CI [+16.1, +34.9]) on anchors—essentially the entire raw effect: what matters is whether the *anchor* leaves the prefix. (An identical-prompt subset bounds generation-batch drift at +3.1 pp, within noise; Table 18, appendix.)

Two methodological corollaries: placebo deletions need their own controls—matching position and length is not enough (the single-control contrast *understated* the anchor’s semantic component by  $\sim 12$  pp and manufactured a spurious restart component)—and the rating-0/+1 “semantic” effects of the single-control analysis (+5.7 to +6.6 pp) vanish under DiD (-1.9 to +1.2 pp, n.s.): target-specific removability lives *only* in the anchor group.

**Paraphrase triangulation.** The placebo isolates *that* the target matters; a semantic-preserving paraphrase isolates *why* (Table 5, appendix). In the anchor group, replacing the erroneous step with an independent-model rewording that preserves its (wrong) meaning leaves accuracy at control level

(+1.7 pp, CI [-6.2, +9.9]), while deleting it gains +19.2 pp over the paraphrase (CI [+8.2, +30.1]): the harm travels with the *semantics*, not the surface form. The fragile-neutral group shows the mirror image: a large raw gain (+24.1 pp) that placebo—and even paraphrase—matches ( $C1-C2 = 0.0$  pp): nothing target-specific, just conditioning on fragile control runs. The stably-correct group’s vs-C0 contrasts inherit the same conditioning bias in the opposite direction (the group is selected on its C0 draws, so regression to the mean pushes  $C1-C0$  and  $C3-C0$  negative); its unconditioned contrasts show deletion  $\approx$  paraphrase ( $C1-C3 = -2.7$  pp, n.s.) and an inert operator ( $C2-C_p = -0.8$  pp), so we do not read the -8.3 pp paraphrase row causally. A blinded 140-item human fidelity audit backs the C3 premise (97.1% faithful or minimally deviating; zero of 46 audited anchor-group paraphrases corrected the original error), and the 19 excluded paraphrases resemble the 221 retained (Table 20, appendix).

Eight fully human-verified cases (78 audited continuations) ship with the released artifacts.

## 7 External Validation on ProcessBench

Does any of this survive outside PRM800K trajectories and outside human step ratings? We sample 300 ProcessBench steps (Zheng et al., 2025)—152 human-annotated *first-error* steps and 148 *locally-correct* steps, balanced across GSM8K, MATH, OlympiadBench, and Omni-MATH (75 each), drawn from a benchmark whose trajectories span twelve source models (six appear in our sample)—and run the identical protocol (3 runs per condition; 204/300 steps admit a length-matched placebo; gold answers matched for 99.5% of records).

The core dissociation transfers. Deleting first-error steps yields +24.8 pp (CI [+16.9, +32.7]); on the placebo-matched subset, the difference-in-differences semantic component is +35.4 pp (CI [+22.8, +48.9])—though here, unlike every no-thinking PRM800K cohort, the own-controlled placebo is itself significantly *negative* (-8.4 pp, CI [-15.9, -1.4]), so part of the DiD reflects a truncation penalty and the raw +24.8 pp is the placebo-free reading. Either way, erroneous steps in other models’ reasoning are negative anchors too. Deleting locally-correct steps yields -2.0 pp raw (CI [-8.3, +4.1]) and a -7.1 pp (n.s.) semantic component: not usefully removable. One



boundary emerges: the DiD semantic component is significant on GSM8K (+37.3 pp) and MATH (+21.1 pp) but attenuates on Olympiad-Bench (+6.0 pp, CI crosses zero) and vanishes on Omni-MATH (−0.6 pp)—and own-controlled placebos show no restart benefit there either (+6.3 pp, n.s.): on problems near the generator’s competence ceiling, nothing target-specific survives. Negative anchors are a phenomenon of *recoverable* trajectories.

## 8 Do Correctness Signals Certify Rollback Safety?

If PRM-style signals suffice to trigger safe rollback, they should predict the two discordance events a policy cares about: *danger* (a correct control draw paired with a wrong deletion draw) and *benefit* (the reverse), both conditioned on the control draw’s outcome. We evaluate step-level signals with problem-grouped 5-fold cross-validation, L2-regularized logistic models, and cluster-bootstrap CIs (Table 12, appendix).

The asymmetry matters. Signals rank *benefit* discordance moderately well: low rating and Harmful type genuinely enrich for rescuable runs. But for *danger* discordance—the safety-relevant direction—the rating alone is anti-informative (AUROC 0.422): among runs whose control draw is correct, low-rated steps are *not* safer to delete. Only features that peek at trajectory state (control stability; effectively extra compute) move danger prediction appreciably. Cheap local correctness is calibrated for “was this step wrong?”, not for “will removing it break the run?”. These AUROC/AUPRC numbers are exploratory discordance diagnostics; the safety conclusions rest on the held-out net effects of Section 9.

**Deployed PRMs, uncertainty, and counterfactual signals.** Because “PRM scores” in deployment are continuous model outputs rather than dataset ratings, we scored the identical 600 steps with three trained PRM checkpoints (Qwen2.5-Math-PRM-7B, Math-Shepherd, RLHFlow-Llama3.1-8B) and two prompted LLM-judge PRMs, plus step entropy/NLL under teacher forcing and a masked answer-probability drop  $I_{\text{mask}} = \log p(y^* | h) - \log p(y^* | h \setminus s_t)$  (Table 13, appendix). The best trained PRM tracks the human rating almost exactly (danger AUROC 0.57 vs. 0.56; step-level, favorable orientation—the run-level, deployment-orientation figure for

the raw rating is the below-chance 0.42 above), and the other two trained checkpoints sit at or below chance (0.47, 0.51)—so the construct gap is not an artifact of auditing labels instead of scores. Token-level uncertainty carries no removability information at all. Two signals stand out modestly: cross-PRM disagreement (benefit AUPRC 0.34), supporting uncertainty-based *abstention*, and the counterfactual mask drop—informative but gold-answer-dependent, hence offline-only.

## 9 From Scores to Policies: Risk–Coverage

If a system triggers rollback on the top- $k$  steps ranked by a signal, does it *certifiably* not lose accuracy, and how many steps does it hurt? A tempting metric—the rate at which control-correct runs come back wrong after deletion—is not causal harm: outcomes are independent samples, so a null deletion on a step with  $p^c = p^t = 0.5$  would show 25% “harm” from sampling disagreement alone. We therefore budget the *certified net accuracy change per rollback*: a policy is certified at budget  $b$  if the one-sided 95% lower confidence bound (problem-cluster bootstrap) of the mean  $\Delta^{\text{tgt}}$  over its selected steps stays above  $-b$ . Alongside, we report step-level *tail risk*: the share of selected steps with observed  $\hat{\Delta}_i < 0$ , minus the exact share expected under the null  $p^t = p^c$  (Binomial(4,  $\hat{p}$ ) per step). Policies: random; correctness rankings (rating-first, Harmful-first, intersection-first); the three trained PRM scores; a deployable trajectory-state predictor (model D) and its outcome-informed variant (model E; Section 8); an in-sample oracle.

On the *balanced* cohort certification is degenerate: the mix is so error-rich (mean effect +7.2 pp, LCB +4.4) that every policy—including random—certifies deleting *all 600 steps*. It becomes meaningful only in the deployment frame (natural chosen-path prevalence, next paragraph), where deleting all traffic has LCB −4.4 pp and certifies nothing below 5 pp.

**Balanced cohort vs. natural prevalence.** The cohort is case–control balanced (200 steps per rating); deployment traffic is not. Our reference frame is the annotator-chosen path of the phase-2 test split—the steps a deployed sampler would carry—comprising 16,481 rated steps, only 31 rated −1 (Table 17, appendix). Chosen paths are error-scarce; our rating−1 targets are drawn from the far larger rated-but-rejected pool (23.2% rated −1) and spliced onto the chosen prefix, so reweight-

| Policy                     | Certifies | Met         | Coverage | Net/del. (pp) | Tail exc. |
|----------------------------|-----------|-------------|----------|---------------|-----------|
| Random                     | 9/20      | 2/9         | 0.00     | −3.7          | 0.043     |
| rating-first               | 20/20     | 15/20       | 0.49     | +0.9          | 0.044     |
| Harmful-first              | 12/20     | 6/12        | 0.20     | −1.8          | 0.064     |
| PRM (Qwen2.5-Math)         | 20/20     | <b>9/20</b> | 0.18     | − <b>6.3</b>  | 0.089     |
| PRM (Math-Shep.)           | 19/20     | 16/19       | 0.33     | +6.4          | 0.000     |
| PRM (RLHFlow)              | 14/20     | 8/14        | 0.05     | +3.3          | 0.038     |
| Predictor (D)              | 15/20     | 13/15       | 0.13     | +20.8         | 0.000     |
| Predictor (E) <sup>†</sup> | 20/20     | 18/20       | 0.58     | +8.8          | 0.010     |

Table 3: Nested policy evaluation (natural prevalence, 1 pp budget, 20 outer 50/50 problem splits): predictors trained on the calibration half with inner 5-fold cross-fitting, coverage chosen there by scanning  $k$  against a *pointwise* cluster LCB (an optimistic screen, not a multiplicity-corrected bound), test scored once. Certifies/Met: nonzero certified coverage / realized test net within budget (even random passes calibration in 9/20); other columns are medians. <sup>†</sup>E reads the control outcome that defines  $\Delta^{\text{tgt}}$  (outcome-informed upper bound, not deployable). Bold: the non-transfer case.

ing assumes spliced alternatives behave like the rare naturally-carried −1 steps and adjusts rating composition only (a joint rating×position reweighting gives −0.3 pp, indistinguishable). Under these weights the average deletion effect moves from +7.2 pp to −0.5 pp (CI [−5.1, +4.0]), with danger and benefit discordance near parity (9.8% vs. 9.3%; Table 16, appendix): in natural traffic, indiscriminate deletion has no upside while the danger does not shrink.

**What certifies in natural traffic.** Under natural weights with in-sample selection (Table 19, appendix), correctness rankings certify only the error tiers at a 1 pp budget: rating-first covers 2.4% of traffic (+9.8 pp per rollback), Harmful-first 3.8% (+17.0 pp), and the best trained PRM score just 0.02% (+51 pp). These are error-correction interventions: the rating-+1 mass (92% of traffic) never certifies below a 5 pp budget.

**Honest nested selection.** Certification must survive selection. Table 3 repeats the exercise with nested train/calibrate/test selection, and three failures emerge. *Instability*: with ∼150 calibration problems the 95% bound is noisy enough that even *random* certifies in 9 of 20 splits, meeting the budget in only 2. *Checkpoint-dependent transfer*: Qwen2.5-Math—the PRM with the best danger diagnostic—certifies in every split yet meets the 1 pp budget in only 9 of 20, realizing −6.3 pp per deletion, while Math-Shepherd transfers in 16 of its 19 certifying splits (+6.4 pp); two PRMs trained for the same construct thus give opposite safety

behavior, and correctness quality does not predict which. *Weak payoff*: rating-first meets the budget in 15 of 20 splits but realizes only +0.9 pp per deletion, and the only broad certifier (predictor E, 58% coverage) earns its coverage by reading the control-run outcome that defines  $\Delta^{\text{tgt}}$ —an outcome-informed upper bound, not a deployable signal, while the honest static predictor D certifies the error tier (13/15 met, +20.8 pp) at only 13% coverage. Because  $k$  is chosen against a pointwise bound and the outer splits share problems, we read the realized test net, not calibration coverage.

### Rollback: one probe run buys most of the safety.

Finally, we simulate the cheapest counterfactual validator: generate *one* probe continuation after a candidate deletion, roll back unless the probe answers correctly, and evaluate on held-out runs. Across all 600 steps this raises the net gain per candidate from +7.2 pp to +11.5 pp (LCB +9.3) and drops the harmed-step share from 12.2% to 3.0%—below its 6.8% stochastic floor, so no detectable excess harm survives—for one extra generation per candidate (63% acceptance). The probe is scored by the same gold-answer judge, so this upper-bounds a one-probe validator and presupposes one; the gain is not mere resampling, since matched truncations that *keep* the target are inert against their own controls (Table 2).

## 10 Discussion and Conclusion

Our results do not indict PRMs at their trained task: low ratings flag wrong steps, where deletion *benefit* concentrates. The failure is direction-specific—correctness scores carry almost no information about deletion *danger*, which is governed by trajectory state and generator recoverability rather than by local correctness. Because the resulting certificates are checkpoint-dependent and unstable across splits, a system that reuses PRM scores to trigger rollback should treat them as candidate nominators, validate each candidate counterfactually, and budget harm explicitly; a single probe-and-rollback generation already recovers most of an oracle’s safety at bounded cost. These conclusions are established for mid-size open generators on mathematical reasoning, and for no-thinking generation in particular—the regime in which negative anchors exist to be removed.



## Limitations

**Generator scope.** Primary results use Qwen3-8B without thinking; replications add phi-4, DeepSeek-R1-Distill-Qwen-14B, and a within-checkpoint thinking-mode toggle (300 steps, 3 runs each). The two no-thinking models agree qualitatively but not step-wise (anchor DiD semantics: +25.4 vs. +34.9 pp; sign agreement 65%); thinking mode on the same checkpoint neutralizes anchors, and the R1 distill replicates the collapse—so anchor-driven gains should not be assumed beyond no-thinking generation. All generators are mid-size open models on mathematical reasoning; we make no claims about frontier models or other domains. The R1-Distill replication could not be run at freeze time (the model was unavailable via the API provider; Section 4) and was completed later on local hardware; phi-4 remains the frozen second generator.

**Case-control cohort.** Rating-balanced sampling (200 per rating) supports subgroup contrasts, not deployment averages: reweighted to the natural prevalence of PRM800K phase-2 test steps (0.2% rated −1), the mean deletion effect is −0.5 pp and anchor steps are rare. Coverage numbers are fractions of the ranked cohort steps under the stated weighting (“traffic” shares are natural-prevalence-weighted), not deployment volumes.

**PRM coverage.** The deployed-PRM audit covers three trained checkpoints—Qwen2.5-Math-PRM-7B, the original Math-Shepherd PRM (Wang et al., 2024), and an RLHFlow Llama3.1-8B PRM trained on DeepSeek rollouts—plus two *prompted* LLM-judge PRMs; trained *generative* PRMs (Khalifa et al., 2025) remain uncovered.

**Semantic label reliability.** Step types come from AI-assisted annotation with human calibration; the two label passes disagree on 27.3% of steps, and the blinded annotation pilot shows only moderate human agreement ( $\kappa = 0.40$ ). The human-only consensus sensitivity analysis (Table 6) reproduces the Harmful-anchor direction (+25.0 pp raw; +22.9 pp legacy placebo-corrected) on 15 exact-consensus Harmful steps, but this is too small to establish a full-cohort human ground truth. Type-based subgroup boundaries are therefore approximate; the pilot covers 120 of 600 steps, and no adjudicated relabeling of the full cohort exists. The headline interaction survives under both frozen label passes and this human-only sensitivity check, but finer type-level claims inherit label noise.

**Judge noise and self-evaluation.** The 93.0%

agreement and  $\leq 0.08$  pp substitution sensitivity are measured on 200 audited Qwen3-8B outputs and do not bound error on the unaudited remainder; phi-4 and ProcessBench continuations are scored by the same judge pipeline without a fresh per-generator human audit. The deterministic-grader regrade (Section 4; Appendix A) bounds self-evaluation bias on the primary cohort only—the extension cohorts have no deterministic regrade.

**Placebo scope.** Matched placebos exist for 511/600 primary steps (position SMD  $\leq 0.298$ ); the external cohort’s placebo matching uses character length ( $\pm 20\%$ ) rather than tokenizer length, and 96/300 external steps admit no match.

**Offline policy evaluation.** Risk-coverage and rollback numbers reuse frozen runs; an online system would face distribution shift from its own interventions. The rollback probe is drawn from the same run pool it is evaluated against, which can flatter acceptance quality slightly. In-sample coverage selection flatters every policy; the nested train/calibrate/test protocol of Section 9 is the honest reading. Even there, the 20 outer splits share problems (each appears in roughly half the test halves), so split counts are dependent, and any policy’s apparent transfer is itself selected on these splits (a winner’s curse); predictor E additionally reads the control-run outcome that defines the target effect, so its coverage is an outcome-informed upper bound, not a deployable certifier. With  $K=4$  runs per condition, per-step harm attribution is noise-limited: the tail-risk excess over the exact stochastic floor is a conservative signal of truly harmed steps, not a per-step identification. Rollback presupposes a verifier and spends one probe generation per candidate, which must be netted against any claimed efficiency savings in deployment.

**Paraphrase validity.** C3 paraphrases passed automatic meaning/length checks (7.9% excluded), and a blinded 140-item human fidelity audit rated 97.1% faithful or minimally deviating, with zero meaning changes among the 46 audited anchor-group paraphrases; two neutral-group paraphrases (4.3% of that group’s audited items) were judged meaning-changed. C3 conclusions rest on this audited premise.

## Ethical Considerations

The study uses the public PRM800K dataset (MIT-licensed problems from MATH) and open-weight

models; no personal data is processed. Beyond the authors, four annotators and one adjudicator—unpaid volunteers from the authors’ research group—labeled reasoning steps and paraphrase fidelity for public math problems (about one hour each); they were informed of the study’s purpose and consented, and the annotation sheets collected no personal information. Counterfactual rollback of reasoning steps is an accuracy/safety intervention; our safety framing (harm budgets) is about answer correctness, not user harm. We release artifacts to support reproduction and auditing.

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## A Judge Audit Detail

|                                                      |          |
|------------------------------------------------------|----------|
| Audited outputs                                      | 200      |
| Binary agreement                                     | 93.0%    |
| Correct-class F1                                     | 0.931    |
| Pair-transition agreement (60 pairs)                 | 90.0%    |
| Max condition bias                                   | 2.3 pp   |
| $\Delta$ overall target effect after substitution    | −0.08 pp |
| $\Delta$ placebo-corrected effect after substitution | −0.05 pp |

Table 4: Judge audit summary. Gates (agreement  $\geq$  90%, bias  $\leq$  5 pp) pass; all headline numbers are insensitive to full substitution of adjudicated labels.

**Deterministic grader.** Because the LLM judge and the primary generator share a checkpoint, we regrade all 6,314 primary runs with a deterministic grader: PRM800K-style answer normalization followed by a SymPy equivalence check. It agrees with the LLM judge on 93.0% of runs, and substituting its verdicts reproduces the overall and anchor deletion effects within 0.9 pp, bounding self-evaluation bias on the primary cohort; the full comparison ships with the audit artifacts.

## B Robustness and Disclosure Tables

**Annotation pilot details.** The blinded pilot re-annotated 120 cohort steps: two independent human labels per step from four annotators in two

| Contrast (pp)                          | Anchor | Stable-corr. | Fragile-neutral |
|----------------------------------------|--------|--------------|-----------------|
| C1—C0 (raw deletion)                   | +20.3  | −11.9        | +24.1           |
| C1—C2 (vs. placebo)                    | +15.6  | +16.8        | +0.0            |
| C3—C0 (paraphrase)                     | +1.7   | −8.3         | +20.9           |
| C1—C3 (vs. paraphrase)                 | +19.2  | −2.7         | +4.1            |
| C2—C <sub>p</sub> (placebo, own ctrl.) | −0.7   | −0.8         | +3.9            |

Table 5: Four-condition strong controls (240 steps  $\times$  4 runs; 80 per group; paraphrase available for 221). Anchors (selected on labels, not outcomes): paraphrase  $\approx$  control while deletion beats both—the *content* is the anchor. Fragile-neutral and Stably-correct are selected on their C0 draws, so their vs-C0 rows are biased by regression to the mean (upward and downward, respectively); the own-control row shows the operator itself is inert. Bold marks contrasts discussed in the text.

pairs, 61.7% raw agreement ( $\kappa = 0.40$ ), with disagreement concentrated on the essential/redundant boundary. Table 6 reports the exact-consensus sensitivity analysis; 93% of consensus-Harmful steps carry rating  $-1$ . An independent adjudicator resolved all 46 disagreements, yielding final labels for all 120 pilot steps, but those adjudications were not used to relabel the full 600-step cohort.

| Human consensus label | $n$ | Rating $-1$ share | Target—Control effect (pp) | Target—Placebo effect (pp) |
|-----------------------|-----|-------------------|----------------------------|----------------------------|
| Essential             | 42  | 21.4%             | +4.2                       | +2.6                       |
| Redundant             | 17  | 11.8%             | −22.1                      | −5.8                       |
| Harmful               | 15  | 93.3%             | +25.0                      | +22.9                      |

Table 6: Human-only semantic-label sensitivity analysis. The 120-step blinded pilot yielded 74 exact-consensus items (46 disagreements omitted from this table); effects are averages over the frozen run-level outcomes for those consensus items. Target—Placebo is the legacy shared-control contrast; the main analysis uses own-control DiD. The Harmful row reproduces the direction and magnitude of the anchor result, but its  $n = 15$  and the pilot’s  $\kappa = 0.40$  make it robustness evidence, not a full-cohort human ground truth.

| Position | Steps | Target—Control (pp) | 95% CI (pp)   |
|----------|-------|---------------------|---------------|
| Early    | 201   | +5.0                | [−1.1, +11.2] |
| Middle   | 201   | +9.5                | [+3.3, +15.6] |
| Late     | 198   | +7.2                | [+1.8, +12.6] |

Table 7: Position-binned target-deletion effects in the primary cohort. Estimates and 95% percentile cluster-bootstrap intervals are from the frozen analysis artifact; the bins were balanced by design. The effect is not confined to early positions.

**Predictor features.** Model D uses 12 static features: indicators rating= 0 and rating= +1; indicators type=Redundant and type=Harmful (initial

| Statistic                           | Observed | Random-label 95% interval |
|-------------------------------------|----------|---------------------------|
| Raw $-1 \times$ Harmful effect (pp) | +23.3    | [+14.0, +30.3]            |
| Within- $-1$ Harmful contrast (pp)  | +12.0    | [−12.9, +13.5]            |
| Calibrated candidate coverage       | 26.7%    | [9.5%, 13.0%]             |
| Calibrated candidate effect (pp)    | +20.9    | [+13.0, +31.1]            |

Table 8: Random semantic-label negative control. Each permutation preserves the initial margins (165/201/234) or calibrated margins (198/199/203), as appropriate, and uses the same 600 frozen outcomes. The within-rating contrast is the mean effect for rating- $-1$  Harmful steps minus rating- $-1$  non-Harmful steps; candidate coverage is the share selected by rating- $-1 \times$  Harmful. Intervals are the 2.5th–97.5th percentiles over 5,000 permutations (seed 20260729). The raw  $-1 \times$  Harmful effect stays positive because the rating- $-1$  effect is retained; the within-rating contrast, centered at zero, is the sharper diagnostic. This supports label enrichment over a frequency-only reading; it is not a claim that the label alone identifies safe rollbacks.

| Selection / contrast                | $n$    | $\Delta^{\text{tgt}}$ (pp) | 95% CI (pp)    |
|-------------------------------------|--------|----------------------------|----------------|
| Rating $-1$ only                    | 200    | +22.0                      | [+15.5, +28.5] |
| Rating $-1 \times$ Harmful          | 178    | +23.3                      | [+16.6, +30.1] |
| Rating $-1 \times$ non-Harmful      | 22     | +11.4                      | [−8.0, +31.8]  |
| Within $-1$ : Harmful $-$ non-Harm. | 178/22 | +12.0                      | [−9.2, +33.0]  |

Table 9: Semantic-label ablation using the primary AI-assisted labels. The rating-only row ignores semantic type; the next two rows partition the same 200 rating- $-1$  steps. CIs are 5,000-replicate step-level cluster bootstraps (seed 20260802). The small non-Harmful subgroup makes the within-rating contrast a resolution check rather than a high-powered standalone test.

pass); their four interactions; position-bin indicators (middle, late);  $\log(1 + \text{target tokens})$ ;  $\log(1 + \text{prefix tokens})$ . Model E adds the control-correct frequency (13 features). Both are L2-regularized logistic models trained with 5-fold cross-validation grouped by problem; every policy score is an out-of-fold prediction. Model E’s added feature is the control-run success frequency, one of the two terms in  $\Delta^{\text{tgt}} = \bar{Y}^t - \bar{Y}^c$ ; problem-level cross-fitting cannot remove this within-step dependence on the evaluated outcome, so E is reported as an outcome-informed (oracle-style) upper bound rather than a deployable predictor. Model D uses no outcome-derived feature.

**Judge configuration.** The judge is Qwen3-8B served at temperature 0, prompted with the problem, the gold answer, and the generated final answer, and instructed to return a JSON verdict on mathematical equivalence; outputs with no extractable final answer are failed by rule. The audit sampled 200 outputs stratified by condition and ver-



| Position     | Group                  | <i>n</i> | $\Delta^{\text{tgt}}$ (pp) |
|--------------|------------------------|----------|----------------------------|
| Early        | Anchor                 | 56       | +16.1 [+4.9, +28.1]        |
| Early        | Other                  | 145      | +0.7 [−6.4, +7.9]          |
| Middle       | Anchor                 | 63       | +24.6 [+13.9, +35.3]       |
| Middle       | Other                  | 138      | +2.5 [−4.3, +9.8]          |
| Late         | Anchor                 | 59       | +28.8 [+16.5, +41.5]       |
| Late         | Other                  | 139      | −2.0 [−7.0, +2.9]          |
| Early–Middle | Interaction difference |          | −6.7 [−25.3, +11.5]        |
| Early–Late   | Interaction difference |          | −15.4 [−33.6, +3.8]        |
| Middle–Late  | Interaction difference |          | −8.7 [−26.9, +10.1]        |

Table 10: Anchor-by-position robustness using the primary initial labels. Anchor = rating  $-1 \times$  Harmful; Other = all remaining steps. Effects are target deletion minus control, with 95% percentile cluster-bootstrap intervals over steps (5,000 replicates, seed 20260801). Interaction differences compare the Anchor–Other contrast between the named positions; every interval includes zero. Position bins are the pre-existing balanced early/middle/late bins, not redefined for this analysis.

| Predictor                        | Coefficient (pp) | 95% CI (pp)   | Reference       |
|----------------------------------|------------------|---------------|-----------------|
| Rating $-1$                      | +17.9            | [+5.3, +30.9] | vs. rating $+1$ |
| Rating 0                         | −0.8             | [−8.7, +7.5]  | vs. rating $+1$ |
| Redundant                        | +3.8             | [−5.1, +12.7] | vs. Essential   |
| Harmful                          | +7.1             | [−4.1, +18.4] | vs. Essential   |
| Middle                           | +3.9             | [−4.5, +12.2] | vs. Early       |
| Late                             | +2.2             | [−5.6, +10.3] | vs. Early       |
| $\log(1 + \text{target tokens})$ | +1.2             | [−4.9, +7.5]  | per log unit    |

Table 11: Length-adjusted deletion-effect regression over the 600-step primary cohort. The outcome is  $100\Delta_i^{\text{tgt}}$ ; predictors are rating indicators, initial AI-assisted semantic-type indicators, position indicators, and log target-token count. Estimates are OLS coefficients with 5,000 problem-cluster bootstrap replicates (303 problem clusters, seed 20260802). The length coefficient is small and its interval includes zero; the Harmful coefficient is also not distinguishable from zero after adjustment.

dict; two annotators labeled blind to condition, with disagreements adjudicated. Full prompts, seeds, and adjudication records ship with the release.

## C Reproducibility

Primary numbers derive from two frozen master tables (600 steps; 6,314 runs) via a dependency-free Python pipeline with fixed seeds (cluster bootstrap seed 20260722, downstream 20260723); a single command regenerates every table, figure source, and the numeric manifest consumed by this paper. The complete accounting is shown in Table 21: 6,314 primary runs plus 17,126 extension runs equals 23,440 generation slots in total. The extension cohorts add 17,126 runs (cross-generator 2,691; long-CoT replication 2,691; thinking-mode toggle 3,582 including its 891 own-controls; strong controls 1,843 = 884 paraphrase C3 + 959 placebo

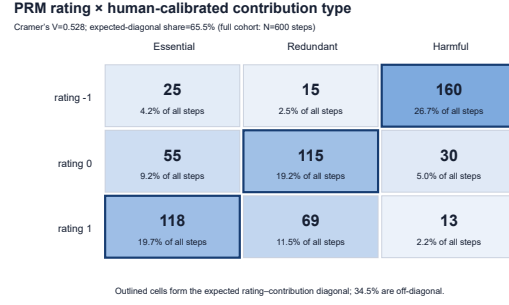


Figure 3: PRM800K rating vs. human-calibrated semantic contribution for the 600-step cohort. The association is strong (Cramér's  $V = 0.528$ ) but 34.5% of steps are off-diagonal: local correctness does not determine contribution.

| Model (features)      | Danger AUPRC | Benefit AUPRC |
|-----------------------|--------------|---------------|
| Base rate             | 0.151        | 0.360         |
| A: rating only        | 0.139        | 0.432         |
| B: type only          | 0.208        | 0.407         |
| C: rating + type      | 0.140        | 0.380         |
| D: static context     | 0.160        | 0.412         |
| E: + trajectory state | <b>0.263</b> | <b>0.478</b>  |

Table 12: Out-of-fold prediction of run-level danger and benefit discordance, conditioned on the control draw: danger =  $P(\text{deletion draw wrong} \mid \text{control draw correct})$ , base rate 15.1%; benefit base rate 36.0%. The rating is *below chance* for danger (AUROC 0.42, CI [0.36, 0.49]); only trajectory-state features (model E, which reads the control-run outcome and so requires extra runs) achieve a meaningful margin (danger AUPRC +0.10 over D).

C2, one of the 960 scheduled C2 generations having failed at the provider; ProcessBench 2,412; placebo own-controls 3,907 across four cohorts), each with its own checkpointed generation, judging, and analysis script under the corresponding workstream directory. Robustness artifacts (label-set re-estimation, placebo-cutoff composition, natural-prevalence reweighting, and the calibration/test policy split) are regenerated by dedicated scripts in the same release. The release includes the intervention builder, matched-placebo selector, judge-audit tooling, policy risk-coverage and rollback code, the 120-step blinded annotation pilot, PRM-scoring adapters, the signal-extraction script, and the StepRem benchmark split with its evaluation script.

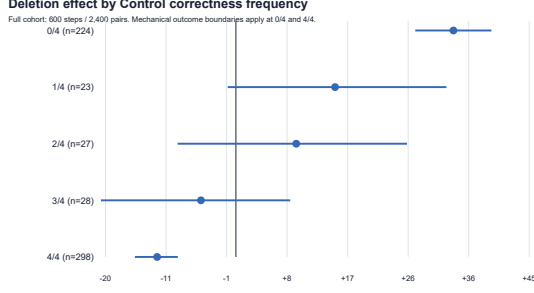


Figure 4: Deletion effect by control-run stability. Gains exist only where control already fails. Outcome-conditioned strata: conditioning on the control draws biases the extreme strata outward (regression to the mean), so these are descriptive bounds, not causal subgroup effects.

| Signal (step-level)         | Danger |       | Benefit     |             |
|-----------------------------|--------|-------|-------------|-------------|
|                             | AUROC  | AUPRC | AUROC       | AUPRC       |
| Base rate                   | —      | 0.15  | —           | 0.24        |
| PRM800K rating              | 0.56   | 0.17  | 0.61        | 0.34        |
| Qwen2.5-Math-PRM-7B         | 0.57   | 0.18  | 0.61        | <b>0.37</b> |
| Math-Shepherd PRM (7B)      | 0.47   | 0.14  | 0.54        | 0.26        |
| RLHFlow PRM (8B)            | 0.51   | 0.15  | 0.59        | 0.28        |
| LLM-judge PRM (Gemini)      | 0.53   | 0.16  | 0.51        | 0.25        |
| LLM-judge PRM (phi-4)       | 0.49   | 0.14  | 0.48        | 0.23        |
| PRM disagreement (5 models) | 0.53   | 0.16  | <b>0.62</b> | 0.34        |
| Step entropy                | 0.46   | 0.15  | 0.48        | 0.24        |
| Step NLL                    | 0.50   | 0.15  | 0.49        | 0.22        |
| Masked answer-prob drop     | 0.56   | 0.19  | 0.57        | 0.30        |

Table 13: Step-level signals vs. deletion outcomes on the identical 600-step cohort (step-level danger/benefit = any control-correct/deletion-wrong, resp. control-wrong/deletion-correct, discordance among the step’s pairs; each signal scored in its favorable orientation, so higher warns against / flags for deletion). The best benefit AUPRCs reach 0.34–0.37 over a 0.24 base—a real lift—while the best danger AUPRC (0.19) barely exceeds its 0.15 base.

| Estimate                                   | Cluster | <i>n</i> | Effect (pp) | 95% CI         |
|--------------------------------------------|---------|----------|-------------|----------------|
| Anchor $\Delta^{tgt}$ , initial            | step    | 178      | +23.31      | [+16.6, +29.9] |
| Anchor $\Delta^{tgt}$ , initial            | problem | 178      | +23.31      | [+16.2, +30.5] |
| Anchor $\Delta^{sem}$ (legacy), initial    | step    | 151      | +13.85      | [+7.0, +20.8]  |
| Anchor $\Delta^{tgt}$ , calibrated         | step    | 160      | +20.94      | [+13.8, +28.1] |
| Anchor $\Delta^{tgt}$ , calibrated         | problem | 160      | +20.94      | [+13.5, +28.4] |
| Anchor $\Delta^{sem}$ (legacy), calibrated | step    | 137      | +14.84      | [+7.5, +22.1]  |
| Overall $\Delta^{tgt}$                     | step    | 600      | +7.21       | [+3.9, +10.5]  |
| Overall $\Delta^{tgt}$                     | problem | 600      | +7.21       | [+3.7, +10.7]  |

Table 14: Label-set and clustering robustness (independent 5,000-replicate bootstrap draws, so intervals differ slightly from Table 1; point estimates identical). The anchor effect survives the calibrated label pass and problem-level clustering. Rows marked *legacy* use the shared-control placebo contrast of Eq. 1, superseded by the own-control difference-in-differences of Table 2 (+25.4 pp); they are retained for comparability with the frozen analysis plan.

| Rating of placebo-cutoff step | Runs | Share |
|-------------------------------|------|-------|
| +1                            | 900  | 59.4% |
| 0                             | 167  | 11.0% |
| unrated                       | 447  | 29.5% |
| −1                            | 0    | 0.0%  |

Table 15: PRM800K ratings at the cutoffs of the 1,514 placebo runs: the placebo never truncates at a known-bad step.

|                             | Balanced cohort     | Natural prevalence |
|-----------------------------|---------------------|--------------------|
| Overall $\Delta^{tgt}$ (pp) | +7.21 [+3.9, +10.7] | −0.52 [−5.1, +4.0] |
| Run danger rate             | 8.5%                | 9.8%               |
| Run benefit rate            | 15.7%               | 9.3%               |
| Dangerous-step share        | 15.2%               | 17.5%              |
| Beneficial-step share       | 23.5%               | 17.6%              |

Table 16: Balanced-cohort vs. natural-prevalence estimates (weights 0.2%/7.4%/92.4% for ratings −1/0/+1, from 16,481 rated phase-2 test steps). Run danger/benefit rates are *joint* discordance rates over all pairs (Section 3); the conditional rates used for prediction (15.1%/36.0%) divide these by the control accuracy (danger) or inaccuracy (benefit). CIs are an independent bootstrap draw.

| Frame                            | <i>n</i> −1 | <i>n</i> 0 | <i>n</i> +1 | Total  |
|----------------------------------|-------------|------------|-------------|--------|
| Phase-2 test, chosen path        | 31          | 1,224      | 15,226      | 16,481 |
| Phase-2 test, all completions    | 6,080       | 2,002      | 18,174      | 26,256 |
| Cohort targets from chosen path  | 2           | 198        | 200         | 400    |
| Cohort targets from alternatives | 198         | 2          | 0           | 200    |

Table 17: Sampling-frame audit (integer counts). The 600 cohort targets are unique steps over 303 problems; every cohort problem is in the phase-2 test split. Rating−1 targets come almost entirely from rated-but-rejected alternative completions spliced onto the chosen prefix; reweighting (Table 16) uses the chosen-path frame.

| Cohort / check                      | $\Delta_{own}^{plc}$ (pp)  | $\Delta_{DiD}^{sem}$ anch. (pp) |
|-------------------------------------|----------------------------|---------------------------------|
| Primary, anchors (151)              | −1.6 [−6.5, +3.2]          | +25.4 [+16.1, +34.9]            |
| phi-4, anchors (100)                | −1.3 [−6.6, +3.8]          | +34.9 [+25.8, +44.7]            |
| Strong-control C2, all groups (240) | +0.8 [−2.7, +4.4]          | —                               |
| Anchor cutoffs before / after       | −5.2 / +4.3                | —                               |
| Batch drift (128 identical prompts) | new − old = +3.1 pp (n.s.) |                                 |

Table 18: Own-control placebo (DiD) robustness. Matched innocuous deletions are inert across the no-thinking PRM800K cohorts shown here; the long-CoT generator instead shows a genuine content-agnostic truncation *benefit* (+8.0 pp, Section 5) and the external first-error subset a truncation *penalty* (−8.4 pp, Section 7). The drift row compares fresh  $C_p$  runs against frozen runs with byte-identical prompts; all  $C_p$  runs are a later batch, so any positive drift lowers  $\Delta_{own}^{plc}$  and raises every DiD one-sidedly—a uniform +3.1 pp adjustment would move the overall DiD +7.5 → +4.4 and the anchor DiD +25.4 → +22.3, changing no conclusion.

| Policy             | Budget 1 pp |                    | Budget 3 pp |                    |
|--------------------|-------------|--------------------|-------------|--------------------|
|                    | Traffic     | Net/del. (LCB), pp | Traffic     | Net/del. (LCB), pp |
| Random             | 0           | —                  | 0           | —                  |
| rating-first       | 2.4%        | +9.8 (+2.0)        | 12.2%       | +1.4 (−1.9)        |
| Harmful-first      | 3.8%        | +17.0 (+1.2)       | 3.8%        | +17.0 (+1.2)       |
| PRM (Qwen2.5-Math) | 0.02%       | +51.3 (+35.5)      | 16.3%       | +8.2 (−1.8)        |
| PRM (Math-Shep.)   | 21.8%       | +9.6 (+3.0)        | 31.3%       | +5.2 (−2.8)        |
| PRM (RLHFlow)      | 40.4%       | +5.0 (−0.6)        | 47.7%       | +3.1 (−2.4)        |
| Predictor (D)      | 66.1%       | +4.2 (−0.8)        | 88.9%       | +1.1 (−2.9)        |
| Predictor (E)      | 68.5%       | +4.5 (−0.1)        | 79.7%       | +1.5 (−2.7)        |
| Oracle             | 96.4%       | +3.2 (−0.2)        | 96.4%       | +3.2 (−0.2)        |

Table 19: In-sample causal certification under natural prevalence: largest traffic coverage whose one-sided 95% cluster LCB of net effect per deletion clears the budget, with the realized net (LCB in parentheses). In-sample  $k$ -selection upper-bounds every policy; the nested evaluation (Table 3) is the honest reading.

|          | $n$ | Anchor | Stable | Neutral | Rating −1 |
|----------|-----|--------|--------|---------|-----------|
| Retained | 221 | 33.0%  | 33.9%  | 33.0%   | 34.8%     |
| Excluded | 19  | 36.8%  | 26.3%  | 36.8%   | 36.8%     |

Table 20: Composition of retained vs. automatically excluded C3 paraphrases: share of each set in the three strong-control groups and in the rating−1 tier (which overlaps the Anchor column); mean target length 38.0 vs. 32.7 tokens.

| Experiment component                             | Steps | Conditions                                      | Generation runs |
|--------------------------------------------------|-------|-------------------------------------------------|-----------------|
| Primary deletion (Qwen3-8B, no-think)            | 600   | $C_0, C_1$ : 4 each                             | 4,800           |
| Primary matched placebo deletion ( $C_2$ )       | 511   | 1–4 matched placebo runs per step               | 1,514           |
| Cross-generator deletion (phi-4)                 | 300   | $C_0, C_1$ : 3 each                             | 1,800           |
| Cross-generator matched placebo ( $C_2$ ; phi-4) | 297   | 3 matched placebo runs per step                 | 891             |
| Long-CoT deletion (R1-Distill)                   | 300   | $C_0, C_1$ : 3 each                             | 1,800           |
| Long-CoT matched placebo ( $C_2$ ; R1-Distill)   | 297   | 3 matched placebo runs per step                 | 891             |
| Thinking-mode toggle core (Qwen3-8B)             | 300   | M2 task set; archived thinking-mode slots       | 2,691           |
| Thinking-mode own controls ( $C_p$ )             | 297   | 3 own-control runs per matched cutoff           | 891             |
| Strong controls (placebo and paraphrase)         | 240   | New $C_2$ : 959; new $C_3$ : 884 completed runs | 1,843           |
| ProcessBench deletion                            | 300   | $C_0, C_1$ : 3 each                             | 1,800           |
| ProcessBench matched placebo ( $C_2$ )           | 204   | 3 matched placebo runs per step                 | 612             |
| M12 own-control $C_p$ (primary)                  | 511   | One own control per matched placebo cutoff      | 1,513           |
| M12 own-control $C_p$ (phi-4)                    | 297   | One own control per matched placebo cutoff      | 891             |
| M12 own-control $C_p$ (R1-Distill)               | 297   | One own control per matched placebo cutoff      | 891             |
| M12 own-control $C_p$ (ProcessBench)             | 204   | One own control per matched placebo cutoff      | 612             |
| <b>Total (no double counting)</b>                | —     | —                                               | <b>23,440</b>   |

Table 21: Experiment accounting with no double counting. “Generation runs” counts archived generation slots used by the frozen analyses.  $C_2$  denotes matched placebo deletion;  $C_p$  denotes its additional own control and is listed separately. Reused control/target runs are counted only in their original component. The primary cohort is 4,800 deletion runs plus 1,514 matched-placebo runs; phi-4 and R1 each have 1,800 deletion runs plus 891 matched-placebo runs; the thinking toggle has 2,691 core slots plus 891 own controls; and ProcessBench has 1,800 deletion runs plus 612 matched-placebo runs. The strong-control row reports the  $959 + 884$  completed new runs after provider/paraphrase attrition.